Manav Batavia

Curriculum Vitae

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Education

Purdue University — Indiana, USA

Candidate for Ph.D. in Mathematics — *Aug 2022 - Present*

- Current GPA: 4.00/4.00.
- Advisors: Prof. Linquan Ma and Prof. Uli Walther.
- Awarded the Certificate of Merit for outstanding classroom instruction.

Indian Institue of Technology Bombay — Mumbai, India

Integrated Master's of Science (B.S.+M.Sc.) in Mathematics — *Jul 2017 – Jun 2022*

- Awarded the Institute Silver medal.
- Cumulative Performance Index (CPI) : 9.82/10.
- Minor in Computer Science and Engineering, Minor CPI: 9.60/10.

Publications and Preprints

4. M. Batavia, ***Vanishing of local cohomology in unramified mixed characteristic***, submitted for publication. [arxiv]
3. M. Batavia, K. Mohana Sundaram, T. Murray, V. Pandey, ***The arithmetic rank of residual intersections of a complete intersection***, submitted for publication. [arxiv]
2. H. Ananthnarayan, M. Batavia, O. Javadekar, ***Syzygies of associated graded modules***, submitted for publication. [arxiv]
1. S. Banerjee, M. Batavia, B. Kane, M. Kyranbay, D. Park, S. Saha, H. C. So, P. Varyani, ***Fermat's polygonal number theorem for repeated generalized polygonal numbers***, Journal of Number Theory, Volume 220, 2021, Pages 163-181. [doi]

Invited Talks

A sharp upper bound on cohomological dimension in unramified mixed characteristic

- (*Upcoming*) Commutative Algebra and Algebraic Geometry Seminar, University of Minnesota, Mar 2026.
- Commutative Algebra Seminar, Purdue Univeristy, Feb 2026.

The arithmetic rank of residual intersections of a complete intersection

- (*Upcoming*) AMS Spring Central Sectional Meeting, North Dakota State University, Apr 2026.
- AMS Fall Central Sectional Meeting, St. Louis University, Oct 2025.
- Commutative Algebra Seminar, University of Utah, Sep 2025.
- Commutative Algebra Seminar, Purdue University, Sep 2025.

When do associated graded modules have pure resolution?

- Poster at Connections Workshop in Commutative Algebra, Simons Laufer Mathematical Institute, Jan 2024.

Conferences/Workshops

(*Upcoming*) AMS Spring Central Sectional Meeting | *North Dakota State University*	Apr 2026
(*Upcoming*) CA+ | *Iowa State University*	Mar 2026
Commutative Algebra and Modular Representation Theory | *IIT Madras*	Dec 2025
AMS Fall Central Sectional Meeting | *University of St. Louis*	Oct 2025
Macaulay2 Workshop | *University of Wisconsin-Madison*	Jul 2025
Singularities in Algebra and Geometry | *Centro de Investigación en Matemáticas*	Jun 2025
Séminaire de Mathématiques Supérieures (SMS) 2025 | *University of Toronto*	Jun 2025

Commutative Algebra in Mixed Characteristic | *University of Nebraska-Lincoln* Mar 2025
Howard Rowlee Lecture 2025 | *University of Nebraska-Lincoln* Mar 2025
Cohomologies of Commutative Algebras | *IIT Madras* Dec 2024
CA+ | *Iowa State University* Mar 2025
UweFest | *University of Notre Dame* Aug 2024
Workshop on Recent Trends in Commutative Algebra | *IIT Bombay* Jun 2024
International Conference on Local Rings and Singularities | *IIT Bombay* Jun 2024
Upcoming Researchers in Commutative Algebra (URiCA) | *University of Nebraska-Lincoln* May 2025
Introductory Workshop: Commutative Algebra | *SLMath, Berkeley* Jan 2024
Connections Workshop: Commutative Algebra | *SLMath, Berkeley* Jan 2024
Workshop on Commutative Algebra and Algebraic Geometry in Prime Characteristics | *ICTP, Trieste* May 2023

Technical Skills

Programming Languages: C++, Python, MATLAB.
Softwares: SageMath, Macaulay2, LaTeX, SolidWorks, AutoCAD, XCircuit.

Academic Achievements

- Department Rank 1 in mathematics at IIT Bombay.
- All India Rank of 174 in IIT JEE Advanced (out of more than 200,000 candidates).
- All India Rank of 8 in the B.Math entrance exam for Indian Statistical Institute (ISI).
- Awarded the Mitacs Globalink scholarship for a research internship in Canada.

Undergraduate Research Experience

Master's Thesis and Further Research Jul 2021 – Dec 2022
Guide: Prof. Ananthnarayan Hariharan, Department of Mathematics *IIT Bombay*

- Proved a necessary and sufficient condition for the existence of a pure resolution of the associated graded module corresponding to a module over a Noetherian local ring.
- Independently proved the Auslander-Buchsbaum-Serre equivalences, and studied Burch's work on constructing ideals with arbitrarily large projective dimension, generated by three or fewer elements.

Betti Tables of Quadratic Ideals in Characteristic Two Jun - Aug 2021
Guide: Prof. Alexandra Seceleanu, Department of Mathematics *University of Nebraska-Lincoln*

This research was done as part of the Polymath Jr. 2021 program.

- Succeeded in resolving all quadratic ideals with three generators in a polynomial ring in three variables over a field of characteristic two. The results attained are easily applicable to larger ideals and larger polynomial rings.
- Presented our results at the Young Mathematicians Conference 2021, organised by The Ohio State University.

Generalization of Fermat's Polygonal Number Theorem May – Jul 2019
Guide: Prof. Ben Kane, Department of Mathematics *The University of Hong Kong*

- Acquired minimal number of terms for which certain weighted sums of generalized polygonal numbers are universal.
- The work done resulted in the aforementioned publication in the Journal of Number Theory.

Teaching Experience

Recipient of the 2026 Certificate of Merit at Purdue University for outstanding classroom instruction.

Instructor of Record | ***Purdue University***

- Algebra Qualifying Exam Prep Class, Summer 2025
- Applied Calculus I, Fall 2024

Recitation/Tutorial Instructor

- Calculus III | Fall 2023, Spring 2024, Summer 2024 and Spring 2026, Purdue University
- Calculus II | Fall 2022 and Spring 2023, Purdue University
- Linear Algebra | Spring 2022, IIT Bombay
- Calculus I | Fall 2019, Fall 2022, IIT Bombay

Service

- Organizer of Ideal Conversations, the student commutative algebra seminar at Purdue University. (Aug 2024 - Present)
- Association for Women in Mathematics (AWM) mentor for graduate students at Purdue University. (Aug 2023 - Present)
- Founder of the Department Academic Mentorship Programme in the Department of Mathematics at IIT Bombay.
- Department Academic Mentor for undergraduate math students at IIT Bombay. (Apr 2019 - May 2022)
- Volunteer for the National Service Scheme (NSS) at IIT Bombay, creating waste awareness and imparting environmental education to underprivileged children. (Aug 2017 - May 2018)